ESP32-S3 Multilingual Voice Assistant System

Hardware Interface Specification & Technical Implementation Guide

Project	ESP32-S3 Push-to-Talk AI Voice Terminal	**Target Subnet**	`10.66.47.x` (Port 5000)
Protocols	I2S RX/TX, SPI, I2C, HTTP POST Stream	**Audio Specs**	16 kHz, 16-bit Mono PCM WAV

1. Complete Hardware Pinout & Wiring Configuration

The system connects an ESP32-S3 host microcontroller to digital audio peripherals, an SPI flash card interface, a character display, and an input button.

Component	Pin Label	ESP32 Pin	Bus / Protocol	Function / Operational Notes
INMP441 Mic (Audio Input)	VDD	3.3V	Power	Strictly 3.3V power (5V damages MEMS capsule)
	GND	GND	Power	Common system ground
	SCK	GPIO 32	I2S0 (BCLK)	Serial Clock for audio sampling
	WS	GPIO 25	I2S0 (LRCLK)	Word Select (Left/Right clock)
	SD	GPIO 33	I2S0 (DOUT)	Serial Data Output (RX stream into ESP32)
	L/R	GND	Control	Tied to GND to output on Left channel
MAX98357A Amp (Audio Output)	Vin	5V / VIN	Power	Connect to 5V rail for maximum speaker wattage
	GND	GND	Power	Common system ground
	BCLK	GPIO 26	I2S1 (BCLK)	Bit Clock for speaker driver
	LRC	GPIO 27	I2S1 (LRCLK)	Left/Right Sync Clock
	DIN	GPIO 22	I2S1 (DIN)	Data In (TX stream from ESP32 to Amp)
HW-125 MicroSD (Storage)	VCC	5V	Power	Fed by onboard 3.3V LDO regulator on breakout
	GND	GND	Power	Common system ground
	CS	GPIO 5	SPI (VSPI)	Chip Select (Active LOW)
	SCK	GPIO 18	SPI (VSPI)	Serial Peripheral Clock
	MISO	GPIO 19	SPI (VSPI)	Master In Slave Out
	MOSI	GPIO 23	SPI (VSPI)	Master Out Slave In
16x2 LCD Display (PCF8574 I2C)	VCC	5V / VIN	Power	Requires true 5V for crystal glass polarization
	GND	GND	Power	Common system ground
	SDA	GPIO 21	I2C Data	Inter-Integrated Circuit Data Line
	SCL	GPIO 15	I2C Clock	Custom remapped SCL pin via GPIO matrix
Push Button (Trigger)	Terminal 1	GPIO 14	Digital IN	Configured with internal pull-up (Active LOW)
	Terminal 2	GND	Power	Connects pin to ground upon button depression

2. Power Decoupling & Capacitor Configurations

To eliminate audio crackle, prevent SD card read/write brownout reboots, and suppress Wi-Fi power transients, install the following passive decoupling capacitors across system rails:

- **3.3V Logic Bus (3.3V to GND):** 22 µF electrolytic capacitor (bulk storage) in parallel with a 0.1 µF (100 nF) ceramic capacitor to absorb ESP32 Wi-Fi peak current spikes.
- **5V Speaker Amplifier Rail (VIN to GND):** 100 µF to 470 µF electrolytic capacitor located directly near MAX98357A power pins to sustain speaker transients without dipping the main system voltage.
- **Button Hardware Debounce (GPIO 14 to GND):** 0.1 µF ceramic capacitor across input terminals to damp switch contact bounce.